FRC | Science Fair | Vacuum Robot

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Brief Overview of Events

FRC Magnolia Regional (Mississippi)
3/18 - 3/21



Started using API keys to grab standard information about teams, as well as more information and input sections

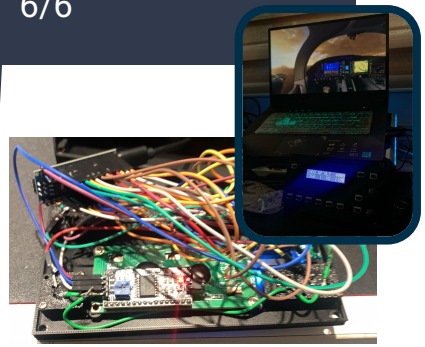
Created the first version of my Pit Collect website and put it to use.



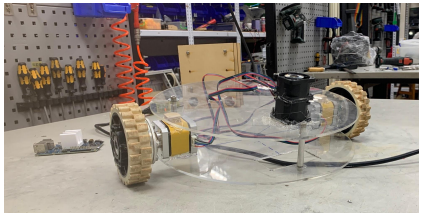
FRC Championship (Houston)
4/29 - 5/1

Built a mini autopilot to control the autopilot in my flight simulator.

Mini Autopilot Panel
6/6



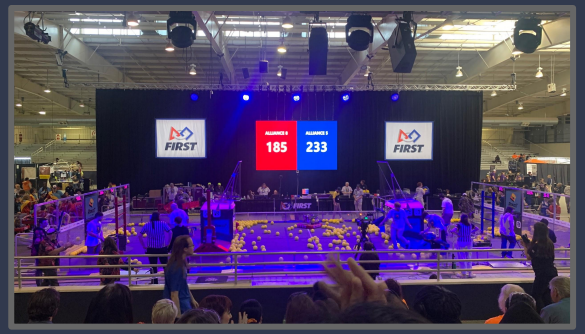
Built a mini autopilot to control the autopilot in my flight simulator.



Vacuum Robot & Science Fair
7/1 - Present


803

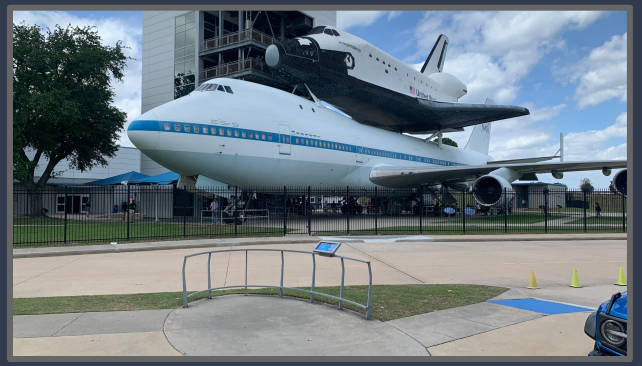

GR
HAAS
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PIRELLI

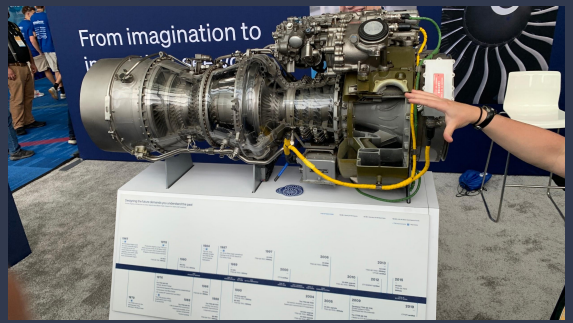

FIRST
ALLIANCE 1
185
ALLIANCE 2
233
FIRST




6998

FRC




From imagination to


FIRST 2026 CHAMPIO

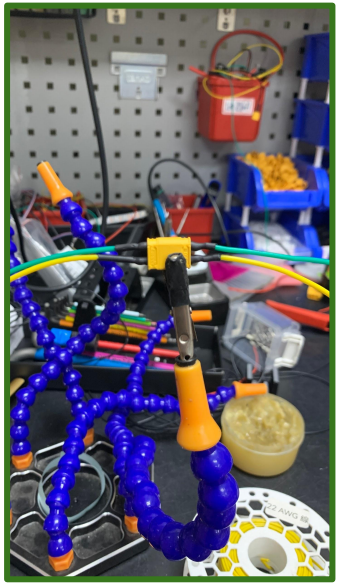
2026 Robot

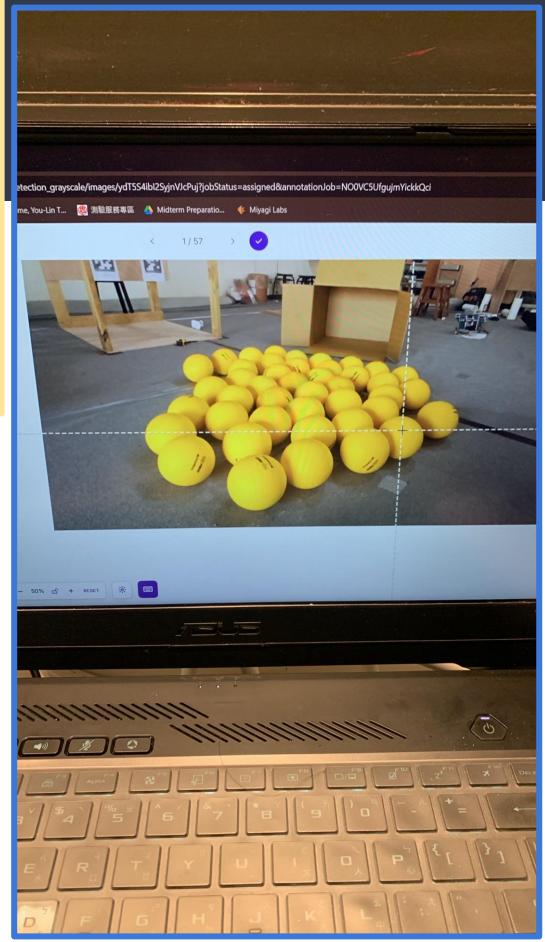


Contributions

- Created a type of scouting website called Pit Collect, which our team used to analyze the hardware and basic strategy of other teams
  - Can be used to figure out how to play against them
  - Determine how to cooperate with them when they are on the same side as us
  - Helped to build an advanced knowledge and understanding of HTML, JS, and CSS through the help of vibe coding and my own code.
- Helped solder part of the wiring for the robot
  - Massively developed my soldering skills
  - Paved the way for future projects
- Worked on object detection models for the 2026 robot
  - Helped to map out game pieces and compile into an object detection model for our robot
  - Taught me the basics of object detection and object detection software, such as Roboflow







FRC Magnolia Regional

3/18 - 3/21



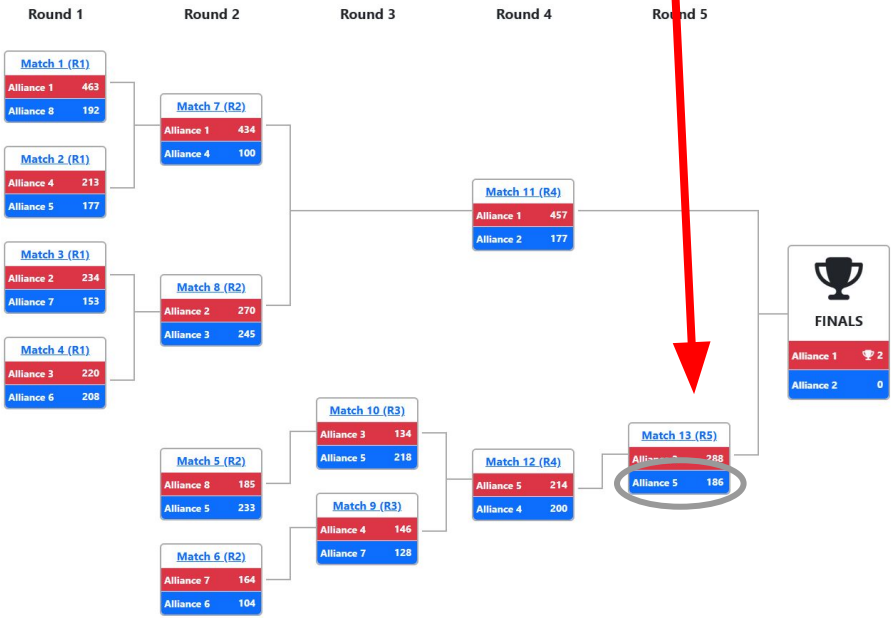
Results



Final Score:
188:288

- 12th out of 46 teams
- Match record of 6-3-0.
- Got into the semifinals of the playoff matches
- Won the Impact Award
  - took us to the FRC Championship in Houston
- My Pit Collect Website was used to track the data of over 40 teams.

Rank	Team	Ranking Score	Avg Match	Avg Auto Fuel	Avg Tower	Record (W-L-T)	DQ	Played	Total Ranking Points*
1	4400	4.11	264.11	40.11	0.00	9-0-0	0	9	37
2	8044	4.00	252.56	53.56	1.11	9-0-0	0	9	36
3	4087	3.33	211.11	44.11	0.00	7-2-0	0	9	30
4	364	3.22	166.89	34.11	0.00	7-2-0	0	9	29
5	2183	3.22	148.78	35.89	0.00	7-2-0	0	9	29
6	2556	3.00	160.56	29.56	0.00	6-3-0	0	9	27
7	3753	3.00	141.78	25.33	0.00	7-2-0	0	9	27
8	2992	2.89	164.78	29.11	1.11	6-3-0	0	9	26
9	3039	2.78	160.56	33.11	0.00	6-3-0	0	9	25
10	1912	2.78	145.22	29.22	0.00	7-2-0	0	9	25
11	6184	2.67	156.33	21.22	0.00	6-3-0	0	9	24
12	6998	2.67	152.56	33.78	0.00	6-3-0	0	9	24
13	9309	2.67	102.00	22.33	0.00	7-2-0	0	9	24
14	10661	2.56	170.00	19.33	0.00	5-4-0	0	9	23
15	4336	2.11	94.33	21.89	0.00	5-4-0	0	9	19
16	456	2.00	124.56	21.33	0.00	4-5-0	0	9	18
17	9503	2.00	108.44	25.56	0.00	4-5-0	0	9	18
18	10101	2.00	81.33	19.33	1.11	5-4-0	0	9	18



Struggles/Problems

QR Code too large

- To transfer data from the device the scouting was done on to the device used to compile all the results, we decided to use a qr code to transfer all the values

- In the early stages of development, the QR code was too large, resulting in the scanner not being able to pick up the QR code

Solution:

Burned Wires

- When I was soldering wires, I sometimes used to accidentally burn the plastic/insulation of the wire and wire connectors, rendering the wires useless.
- The solder would not melt properly, and so I would keep the soldering iron on the component for a long time.

Solution:

QR Code Fix

To solve the issue of my QR code being too large, I utilized 4 main methods:

1. Setting Visual Dimension Limits
   a. Set the QR code size to 256 by 256 px
      i. Ensures the image size will not change
2. LZ-string Compression
   a. Before any of the qr codes are generated, all the TSV(tab separated values) are compressed into Base64 format
      i. E.g. A -> 000000, B -> 000001…
3. Character Sanitization
   a. Newlines and tabs are deleted from text inputs to save space

```
new QRCode(qrEl, {
    text: compressed, width: 256, height: 256,
    colorDark: "#000000", colorLight: "#ffffff",
    correctLevel: QRCode.CorrectLevel.L
});
```

```
const compressed = LZString.compressToBase64(tsvValues.join('\t'));
```

Doesn't this make the file size larger?
It does, but LZ string compression looks for recurring patterns and creates a dictionary for where the values should go

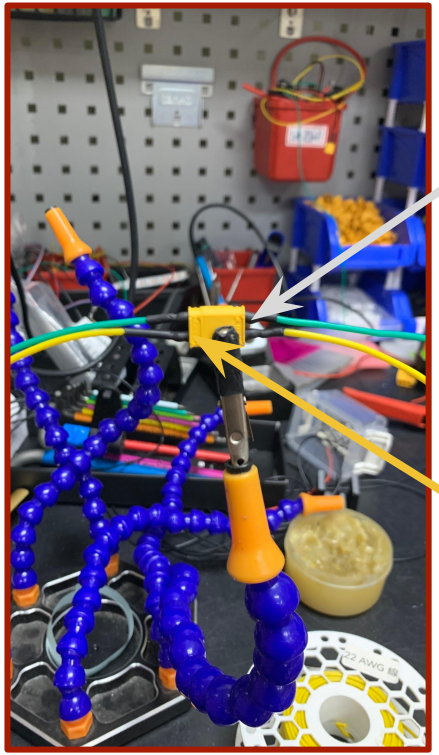
Back:

```
document.getElementById('notes').value.replace(/[\t\n]/g, ' ')
```

Soldering Skills Improvement

Since I kept the soldering iron on the component for too long, it caused the plastic surrounding the metal contacts to melt. This results in the component becoming unusable, which wastes time and resources.

1. Plug a spare connector in
    a. By plugging in a spare connector, it acts as a heat sink. The extra metal contacts mean that more heat is required to raise the temperature of the component, allowing me to keep the soldering iron on for longer.
2. Pre-tin both the connector and the wire
    a. I used to only pre-tin the wire, which meant that there will be poor heat transfer
    b. By tinning the connector too, the metal heats up way quicker



Back:

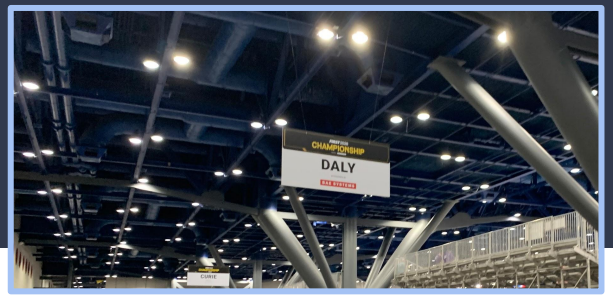
FRC Houston Championships

4/29 - 5/1



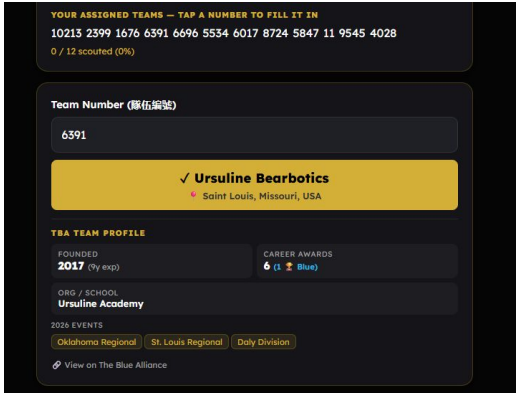


Results



45		The Pascack PI-oneers	1.80	381.40	80.30	2.50	4 - 6 - 0	10
46	624	CRyptonite	1.80	361.60	74.50	4.00	4 - 6 - 0	10
47	4189	Chargers	1.80	360.90	74.10	4.50	4 - 6 - 0	10
48	2718	Team OKC e'possums (Σ 1/n! x 10^9)	1.80	351.00	73.80	1.00	4 - 6 - 0	10
49	6998	Unipards	1.80	340.80	62.20	0.00	4 - 6 - 0	10
50	5550	BRONChO BOTS	1.80	296.90	61.40	0.00	4 - 6 - 0	10
51	3880	Tiki Techs	1.70	290.80	60.40	1.00	5 - 5 - 0	10
52	1156	Under Control	1.60	387.90	73.80	1.50	3 - 6 - 0	10
53	1629	Garrett Coalition (GaCo)	1.60	350.90	69.10	2.50	4 - 6 - 0	10

- 9 divisions in total
- 49th out of 75 teams in the Daly Division
- My Pit collect website was used to scout over 70 international teams
- Win loss record of 4-6-0
- Added major features to my Pit Collect website
  - API key grabbing from the Blue Alliance to display basic information about a team.
  - Allowed for teams to draw out multiple autonomous strategies





Struggles/Problems



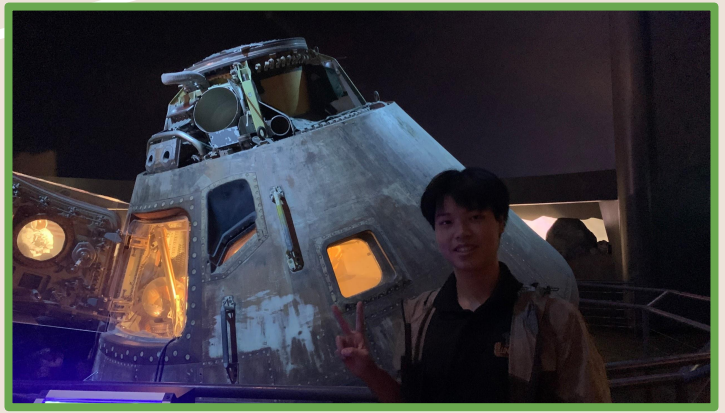
1. Robot wiring failures during match
    a. During a lot of the matches, our robot could not move due to electrical wiring mistakes and zeroing issues due to human error
    b. Time constraints
2. Autonomous path drawing is laggy
    a. To lower the QR code size, I decided to lower the amount of points(x,y) that will be plotted when you drew on the field.
    b. This made the line drawing inaccurate and laggy, which meant we were compromising our data for file size
    c. = lossy compression

1. Although I was not in the fixing team, this experience has taught me that even under pressure, you cannot skip steps in the standard procedure, and must follow the checklist as intended.
2. To fix the

Johnson Space Center











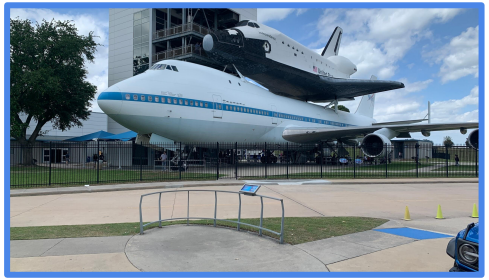


Saturn V Rocket

When I visited the Johnson Space Center in Houston, the thing that stood out to me the most was the Saturn V Rocket. When I approached it, I was in awe of its massive size and complex engineering.

Realization: True engineering requires everything to be done step by step, and everything has to be done to perfection.

Space Shuttle Carrier

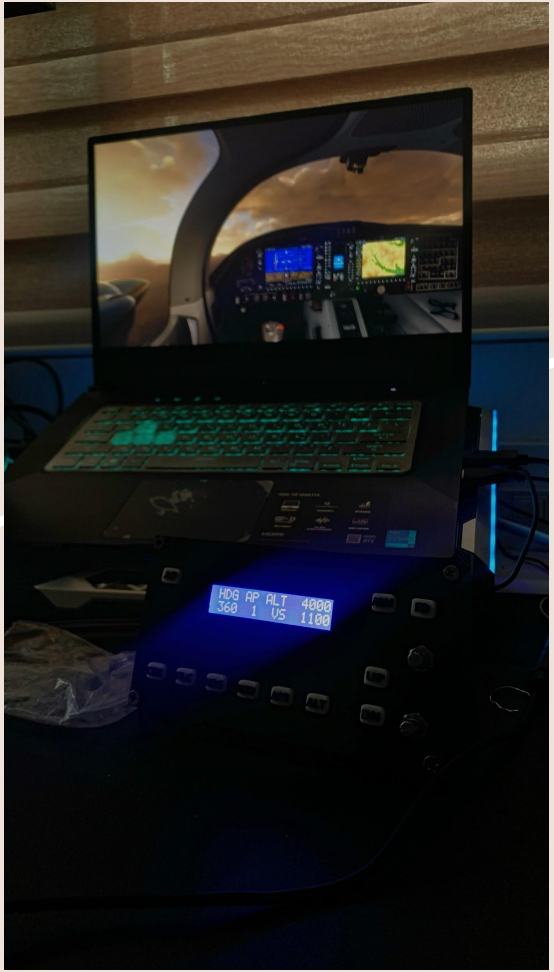








Flight Simulator Autopilot



Autonomous Vacuum Cleaner

Science Fair